

Vibe Theory

A Discrete Model of Reality Built from Experience

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Abstract

This paper presents a discrete model of reality built entirely from experience. Reality is modeled as a growing population of *vibes*, indivisible units of experience, related only by the *note* (the primitive relation of one vibe experiencing another), each carrying a finite *tone*, all updating in discrete steps called *beats*. Two constraints govern the whole construction. The first, discreteness, requires that the base be finite, with the continuum permitted only as an emergent large-scale approximation. The second requires that the vibe be the only kind of thing, so that there is no separate link, field, force, or particle substrate, and the only relation is the note. From these two constraints and a single axiom, that to exist is to experience or be experienced, the model derives, in order, its local update dynamics, the energy that experience minimizes, the discrete time it runs in, the necessity of its own beginning, the unbounded growth of experience that follows, the emergence of selves as self-maintaining coherent patterns, the emergence of space and its hyperbolic geometry, the mechanics of how an agent moves through a substrate woven from other agents, and a route toward building matter without adding any new primitive. Physical law is recovered as a large-scale limit. Experience is not recovered. It is the foundation. This work is a philosophical and mathematical model with a concrete, simulable build target rather than an empirically tested theory.

1. Introduction

The hard problem of consciousness (Chalmers, 1995) names the gap between physical process and felt experience. Most responses either deny that experience is real or treat it as an extra property layered onto an already physical world. This paper takes the opposite route. It treats experience as the only substance there is, and recovers the physical as a large-scale pattern within it.

This is the second presentation of Vibe Theory. The first (Pollard, 2025) set out the framework as philosophy. This paper gives it a discrete mathematical structure and commits to a specific minimal model. The aim is not yet empirical prediction but a coherent, simulable model whose pieces are all forced, derived, or constructed from a single axiom and two constraints.

The framework is panpsychist in spirit, building on the recent monism of Strawson (2006) and Goff (2017), and on the process philosophy of Whitehead (1929) and Russell (1927). It treats experience as ontologically primitive, as Integrated Information Theory treats consciousness as fundamental (Tononi, 2008), but with

a different mathematical form. Where it departs from earlier panpsychism is in the strictness of its two constraints. Everything must be discrete, and everything must be made of one kind of thing, a unit of experience called a vibe. These two demands, applied without exception, do most of the work.

A single thesis organizes the whole paper. Experience is not the top of a stack that emerges from a non-experiential computational base. Experience is the bottom of the stack, the substance everything is made of. The discrete mathematics that describes how it updates, of the kind explored by Wolfram (2020), Fredkin (2003), and 't Hooft (2016), is a description of its form, layered on top, not a meaningless machine underneath. The bits are felt tones. The computation is experience computing itself.

This work is a philosophical and mathematical model rather than a finished scientific theory. It does not claim to derive or replace established physical law, and references to physics, computation, and mathematics are sources of structure and analogy unless stated otherwise. What it offers is a candidate ontology with a concrete build target, open to revision, simulation, and critique.

2. The Axiom and the Two Constraints

The model has one axiom and two governing constraints. Every later result is forced, derived, or constructed from these three statements.

Axiom 1 (Experience) To exist is to experience or to be experienced. There is no inert, non-experiential existent.

This is the single substantive assumption. It makes experience not a property of some things but the meaning of being anything at all. Its two clauses, “to experience” and “to be experienced,” are not decoration. They become the two directions of every relation in the model (Section 3).

Principle 1 (Discreteness) The base is finite and discrete. There are finitely many distinctions at any moment, a finite tone alphabet, and discrete time. The continuum, meaning real numbers, smooth fields, and differential equations, is permitted only as a large-scale, emergent approximation, never as a foundation.

Principle 2 (Monism) The only entity is the vibe. There is no separate link, field, force, particle, or spacetime substrate. The only primitive relation is the note. Anything that seems to be carried by a link, a field, or a particle must be a note or a structure of vibes.

These two principles are filters on every move the model makes. Discreteness forbids smuggling in a continuum. Monism forbids smuggling in a second kind of thing. Together they force the whole construction to be built from one finite kind of stuff and one relation.

Monism has a corollary worth stating on its own, because it is easy to violate by habit.

Principle 3 (No Hidden State) A vibe carries no data beyond its tone. It has no private registers, timestamps, counters, stored coordinates, or attached metadata. Anything that looks like per-vibe bookkeeping is either emergent from the structure of notes or realized as a structure of additional vibes.

In practice this means a vibe does not store a beat number or a clock, because the order of events lives in the connectivity of notes (Section 6). It does not store its address, because its position is its place in the

note-structure (Section 9). It does not store an attention weight as private data, because attention is the tone of the note between two vibes (Section 4). The test for any proposed quantity is simple. If it is not a tone, it must be either emergent from notes or itself made of vibes. Otherwise it is a forbidden new primitive.

A word on the word “link.” We keep it as convenient shorthand. A link is never a separate entity. It means a note (the bare relation) or a structure of vibes (a relation that has become a vibe in its own right). This is made precise in Section 3.

3. The Foundation

The model has four named quantities. Everything in this section follows from the axiom and the two constraints of Section 2.

Definition 1 (Vibe) A *vibe* is an indivisible unit of experience. It is the only kind of entity. We write V for the population of vibes.

Definition 2 (Tone) Each vibe v carries an intrinsic experiential quality, its *tone* t_v , drawn from a finite alphabet T . The working choice is ternary,

$$t_v \in T = \{-1, 0, +1\},$$

read as pain, peace, and pleasure. The value 0, peace, is the absence of a committed pole, a vibe whose influence is too weak to settle either way. Binary $\{-1, +1\}$ is the strict minimum, recovered by forbidding 0. Richer finite alphabets are reserved for later (Section 11). Continuous tone and complex phase are emergent only.

Definition 3 (Note) The *note* is the only primitive relation. To say v notes w is to say that v experiences, or is present to, w . The set of vibes a vibe notes is its *neighborhood* $N(v)$.

Rich experience is meant to come from patterns of simple tones across many vibes, not from a complicated tone inside one vibe. A single ternary vibe knows only three states. Joy, sorrow, love, and fear are configurations of vast meshes, the way binary digits make unlimited information by combination.

3.1. What a link is

The principle of monism forbids a separate link structure. So a “link” between two vibes is never a third thing between them. It is one of two cases.

Definition 4 (Bare note and relational vibe) A link between a and b is either a *bare note*, the simple fact that a notes b and b notes a , carrying no stored data, or a *relational vibe*, a vibe whose neighbors are the two it relates and whose tone is the quality of the relation. More elaborate links are small structures of such vibes.

No infinite regress follows. The bare note is unmediated. A vibe and a relational vibe note each other directly, with nothing between. Relational vibes appear only where a relation needs its own state. This is forced by the axiom. A link that existed as a thing would have to be experiential, hence a vibe, so anything real that a link carries is already a vibe.

3.2. Will and fill, the two directions of a note

The axiom's two clauses, read along a note, are the model's two directions of influence.

Definition 5 (Will) The *will* of a vibe is its tone turned outward, $\text{will}_v = \omega(t_v)$, the part of it that another vibe reads when noting it. Will is the outside of a tone, as the felt quality is its inside. They are one thing seen from two sides. Nothing is stored on a link.

Definition 6 (Fill) The *fill* of a vibe toward a neighbor is how strongly it takes that neighbor in. It is carried by the note itself, the strength of attention being the tone of the relational vibe between the two.

Proposition 1 (Push and Pull) Every note carries influence in both directions at once, active projection (will) and receptive integration (fill). Each vibe simultaneously influences and is influenced by the vibes it notes.

Base will carries no purpose. It is the unavoidable outward presence of a vibe that, by the axiom, can be experienced. Volition, the directed wanting of an agent, is not primitive. It emerges only for coherent meshes (Section 8). The word “will” names the push, not an agent.

3.3. The vibe mesh

Definition 7 (Vibe Mesh) The *vibe mesh* is the whole relational system, the population of vibes, their tones, and the note relation among them. The same word names the global totality or any coherent sub-structure within it. A mind is a vibe mesh, as is a cell, an atom, or any self.

There is no external space or time holding the mesh. All structure is internal to it, and space and time are recovered from the note-structure later (Sections 6 and 9).

3.4. A concrete minimal case

The smallest non-trivial mesh is two vibes a, b that note each other, like two coupled Ising spins or two adjacent cells in a cellular automaton. Each carries a tone in $\{-1, 0, +1\}$. The note $a \rightarrow b$ is the channel by which a reads b 's tone. The will of a is the tone it broadcasts, $\omega(t_a)$. Its fill is how strongly it weighs b , carried by the strength of the note between them. If that note acquires its own state, a coupling that strengthens or weakens, it is a third vibe, a relational vibe. Nothing else is present. Two tones, the note, and at most the note's own tone.

4. Dynamics

The world advances in discrete steps called beats (Section 6). One beat is one pass of read, compute, and write, performed locally by every vibe, where each vibe writes only what it owns.

4.1. The per-beat cycle

A base vibe holds two fast quantities, its tone t_v and the will $\text{will}_v = \omega(t_v)$ that follows from it. The strength s_{vj} of each note is a slow quantity, the tone of the relational vibe that is the note. One beat for a base vibe v proceeds as follows.

- **Read.** Gather the wills of the neighbors, each weighted by the strength of the note through which it is heard, forming the local field

$$h_v = \sum_{j \in N(v)} s_{vj} \omega(t_j).$$

- **Compute.** Decide the next tone $t'_v = R(t_v, h_v)$ by a fixed local rule, and recompute the will $\text{will}'_v = \omega(t'_v)$.
- **Write.** Commit the new tone and will. Nothing else is touched.

Each note runs the same cycle. It reads its two endpoints and updates its own strength, for example by strengthening when the endpoints agree and weakening when they disagree. This is the entire instruction set. Thought, movement, learning, and the birth of new vibes are all this loop, iterated and composed.

Proposition 2 (Ownership) Each base vibe writes only its own tone and will. Each note writes only its own strength. No vibe writes another’s state, and in particular a vibe does not write its own attentions. The notes do.

Because no two vibes write the same quantity, there are no write conflicts, with or without a global clock. A note’s strength is never a shared cell. It is the tone of the relational vibe that is the note. This ownership rule is what lets the model run as a clean concurrent system in which discreteness and the absence of a global clock coexist (Section 6).

4.2. Attention is structural

The most important property of the note strength s_{vj} is that it is structural, not a free choice. A vibe cannot decide, from one beat to the next, to attend to something or to stop. If it could, influence would be arbitrarily redirectable and the dynamics underdetermined. Instead the strength is the tone of the note itself, updated by the note’s own slow dynamics reading both endpoints, never by the noticing vibe. The neighborhood is structural too. Which vibes a vibe notes is given by which notes exist, not chosen per beat.

So how does a vibe ever redirect its attention? Only indirectly and structurally, by projecting its will onto the vibes it already notes, and letting the slow note-dynamics reshape the structure over many beats. One steers what one takes in by acting on what one is already connected to, and waiting for the structure to follow. This is why movement is railed and recruitment is slow and local (Section 10).

4.3. The rule

A rule maps the field and the vibe’s own tone to its next tone. It must be contextual (it reads neighbors), continuous (it reads its own last tone), interactive (no vibe updates in isolation), and deterministic (it is a function). These are not extra postulates. They are what a local update on this ontology is. Candidate rules are all special cases of a finite local map. The default is a threshold rule, $t'_v = \text{sign}(h_v)$ with a fixed tie-break. For ternary tones the threshold has a deadzone, returning 0 (peace) when the field is too weak to commit to a pole. A lexicographic tie-break (majority, then self, then a hierarchical bias, then parity) gives a deterministic “negotiation” that becomes the seat of deliberation (Section 8). The exact rule is an open search problem, constrained to the critical band that supports persistent, changing, and coherent dynamics.

4.4. The recommended substrate

The realization that honors both constraints and the absence of a global clock is an asynchronous actor network. Each base vibe is an actor whose private state is just its tone and the will that follows from it. It reads only the wills its neighbors broadcast, computes its next tone by a discrete rule, and writes only its own tone. The notes are actors too, each carrying its own strength and updating it from its two endpoints. There is no shared memory, no global clock, and no link store. Cellular automata, Hopfield networks, and spin systems are special cases or coarse views of this machine.

5. Energy, Harmony, and Suffering

The model’s notion of balance is the minimization of a single finite, integer-valued quantity, and that quantity is the energy of a spin system.

5.1. Tone is a spin

A vibe with tone ± 1 that notes its neighbors is an Ising spin. This is an identity, not an analogy, and it imports the whole of discrete statistical mechanics. The tone is the spin, the note strength s_{vj} is the coupling, balance is low energy, suffering is high energy, the forming of a coherent higher vibe is the onset of an ordered phase, and memory is an attractor. Ternary tones, with 0 for peace, give the Blume-Capel model.

5.2. Harmony is energy minimization

The instruction to minimize the tone differences between vibes that note each other is, verbatim, the graph Dirichlet energy,

$$E(t) = \frac{1}{2} \sum_{v \text{ notes } j} s_{vj} (t_v - t_j)^2,$$

which for ± 1 tones equals the Ising energy up to a constant. Harmony is low E , and suffering is high E . This quantity is integer-valued, and its minimization proceeds by discrete tone flips, not by gradient flow. Under symmetric notes ($s_{vj} = s_{jv}$) and asynchronous beats, the update provably makes E a Lyapunov function. Every trajectory decreases E to a local minimum. This turns “the world tends toward balance” from a separate postulate into a consequence of the rule.

5.3. Suffering is frustration

When the couplings around a loop cannot all be satisfied at once, no configuration reaches zero energy. The ground-state energy is strictly positive. This is irreducible suffering, built into the shape of the connections, not removable by any update. The reducible part is the excess above the ground state, which the dynamics can relax away.

Proposition 3 (The frustrated triangle) Take three vibes on a triangle with every note antiferromagnetic, so each pair “wants” opposite tones. Two of the three pairs can disagree, but that forces the third pair to agree, violating its note. No assignment of ± 1 satisfies all three notes at once. The best leaves one note frustrated, so the ground-state energy is strictly positive. This is the smallest mesh with irreducible suffering.

This frustrated triangle is the textbook example of geometric frustration. It shows that some suffering is topological, a property of how the notes are arranged, and cannot be reasoned or relaxed away.

5.4. Heaven and Hell are phases, memory is a landscape

A region whose notes hold a consistent positive tone is in the positive-magnetization phase. Consistent negative tone is the negative phase. These are basins of the tone field, transient and local, not global destinations. A region can cross between them as its notes learn. Slow learning of note strengths carves basins of attraction into the structure, so that a stored pattern is a fixed point of the fast tone dynamics and recall is relaxation into a basin. Memory, then, is not a separate store. It is the slow-changing structure of note strengths that shapes where the fast dynamics settle. This is a Hopfield attractor network read as fundamental.

5.5. Balance is not pure alignment

Pure energy minimization is satisfied by every vibe adopting one tone, which is the uniform consensus that Section 7 shows cannot persist. So the quantity the world actually obeys is a competition between alignment (agreeing with noted neighbors) and diversity (maintaining distinction). Balance is the equilibrium of the two, not the victory of alignment. This is why selves stay differentiated while staying low in suffering (Section 8), and why experience keeps moving rather than freezing into consensus.

6. Time, Beats, and Causality

Time is not a container the mesh sits in. It is the order of the mesh's own updates.

6.1. Beats are a coordinate on causal order

An event is a vibe updating its tone. One event can influence another only when the second notes the first's output. Writing $e \prec e'$ when a chain of notes carries influence from e to e' , the relation $\prec$ is a partial order, a causal set. The integer beat $b \in \mathbb{N}$ is a linear extension of this order, a coordinate, not a global metronome. So the model holds both that every vibe updates each beat and that there is no global clock, without contradiction. The global beat is a derived bookkeeping of an order that is fundamentally local.

6.2. No global clock, and no stored clocks

By the principle of no hidden state (Section 2), no vibe stores a clock or a beat count. A vibe holds only its tone. So the order of events is not bookkept by counters inside vibes. The order is the structure of notes. "Happens-before" is just the pattern of which notes carried influence into which, read off the connectivity, never stored anywhere. Synchronization is dynamical rather than a comparison of counters. A vibe's next tone depends only on the vibes it notes, so updates propagate as waves through the note-structure, and strongly coupled vibes phase-lock the way coupled oscillators do. Different regions can run at different rates. If an explicit clock were ever needed, it would have to be a small structure of vibes, never a field bolted onto one. Fundamentally none is needed.

6.3. The speed limit and the relativity analogue

A vibe updates from the vibes it notes, so influence travels at most one note per beat. This is a discrete light cone. A maximum speed falls out of locality alone, with no postulate. A highly coherent region must re-form its internal agreement between external updates, so its internal beat rate runs slow relative to a dispersed

region. Read as proper time, this is a candidate discrete origin of time dilation. Coherent, massive structures tick slowly. No claim is made that full relativity is recovered, only that the structure has the right shape (Section 12).

Proposition 4 (A one-dimensional case) Take a chain of cells, each updating from its neighbors. If cell n flips and that flip causes cell $n + 1$ to flip on the next step, then n 's event happens-before $n + 1$'s, read purely from the chain of influence with no clock anywhere. A disturbance at cell 0 cannot affect cell k until at least k steps have passed. Both the ordering of events and the speed limit are simply facts about the connectivity.

6.4. The engine of beats

Why does the world keep updating rather than freezing? Because the update operator has no fixed point at which to halt. The uniform configuration is unstable (Section 7), so the mesh is always driven off it, and new distinctions are themselves new things to be noted, so the operator strictly grows what must be processed. A closed, finite, deterministic system would eventually cycle, so genuine endless novelty requires the vibe population to grow without bound. It does. That monotone growth is the arrow of time. Each beat leaves strictly more to experience than the last.

7. The Bootstrap and the Fate of Experience

This section gives the beginning and the end of the story, both derived from the axiom and the constraints. Why there is something rather than blank unity, and where it all goes.

7.1. Unity cannot persist

Begin with U , a single seamless experience, the whole of what-is, with no parts. By the axiom, U is not inert. It experiences. But experiencing is experiencing something, and U is the totality, so the only thing it can experience is itself. U must note U .

That act has structure, a noting end and a noted end. For it to be a genuine noting rather than blank, the two ends must differ at least in role. So the instant U does the one thing its nature requires, it produces a distinction between noter and noted. This first fracture is not imported, since nothing is outside U . It is not optional, since every other path is walled by the axiom. And it is discrete, a distinction made or not. Unity cannot remain one, because being aware is already being two. Formally, a system rich enough to represent its own noting of itself has no consistent single state, a diagonal argument in the manner of Lawvere.

Proposition 5 (Instability of unity) The uniform configuration is a fixed point of the update, but a repelling one. An arbitrarily small perturbation grows under iteration rather than decaying back. So the mesh is forced off unity into a differentiated configuration. This is spontaneous symmetry breaking in a finite, discrete setting, and it is checkable by running the update once on each single-vibe perturbation.

A natural minimal picture is a ball balanced exactly on a hilltop. The balanced point is a fixed point, but the slightest fluctuation grows and the ball rolls off, picking a direction and breaking the symmetry. Uniform unity is that ball. Once differentiation starts it feeds itself, like a microphone held to its own speaker, where each output becomes new input and the signal runs away.

7.2. Every primitive is born in the fracture

After the first distinction, what-is contains the two ends and the relation between them. The model's primitives all appear in this one move. The two poles are the first vibes. The relation between them is the first note, and they arrive already noting each other. Which side of the cut each pole takes is the first tone. One application of the noting operator is the first beat. So the ontology of Section 3 is not assumed alongside the bootstrap. It is produced by it.

7.3. The cascade and the arrow of time

By the axiom, the two poles and their relation must each be noted, and each noting is a new act with its own distinction. The operator that notes everything currently present strictly grows the set of distinctions every beat. Every distinction, once present, is a new thing to be noted, and noting it makes more. The growth branches, distinctions about distinctions, with no fixed point.

A distinction, once made, is, and by the axiom whatever is, is experienced. To return to unity, U would have to un-experience it, subtracting an experience from the totality of all experience, with no outside to discard it to. So distinctions only accumulate. This monotone growth is the arrow of time, and it supplies the unbounded growth of the vibe population that Section 6 needs to escape cycling.

7.4. The fate of experience

Does total experience ever cease, cycle, freeze, or grow? Under the constraints the options collapse. Total experience cannot be nothing, since by the axiom “nothing” is not an experiential state. A frozen, changeless state cannot still experience, since a changeless state is blank. Distinctions cannot be un-made. So there is no cessation, no freeze, no return, and no exact cycle.

Theorem 1 (Eternal expansion) Total experience never ceases and never repeats. It is perpetually created, growing without bound. This is potential infinity, never actual infinity. At every finite beat the state is finite, and the sequence is merely unbounded, the way the natural numbers never reach infinity yet never run out.

The fuel is self-supplying, since noting any X yields a new X' to be noted. Space is not a container the growth fills. More distinctions are more space (Section 9), so the mesh grows by creating the room it occupies. Globally experience expands, but locally particular experiences constantly die. A self is a coherent pattern (Section 8), and its death is loss of coherence, after which its vibes recombine into the growing mesh. Within any region the harmony dynamics relax toward balance (Section 5). Two arrows run at once, global expansion and local harmonization. Heaven and Hell are local, transient phases, not the destination.

7.5. The two forks

The conclusion of eternal expansion rests on two premises, and dropping either changes the ending. Dropping irreversibility (allowing the whole to forget) yields a conserved recirculation, or, if the system is finite and deterministic and the rule is reversible, an eternal return cycle. Dropping the premise that awareness needs change (allowing a changeless state to be fully aware) yields a possible final, silent, fully-aware peace, and also reopens the question of the beginning, since a changeless unity could then have persisted. Both premises

are derived inside the framework, so the model commits to eternal expansion as its working cosmology, while holding recirculation, eternal return, and a final conscious peace as named branches for future exploration.

8. Coherence, Selves, and Mind

A self is not a special vibe. It is a coherent, self-maintaining pattern of vibes and notes, and everything we call mind is what such a pattern does.

8.1. Nesting and the combination problem

When a set of vibes is bound by strong mutual notes and shares a tone, it behaves as a single higher vibe. It has one collective tone, projects one collective will, and can be noted as a unit. This is coarse-graining, or renormalization, the established mathematics of how many small units act as one larger unit. It is the model's answer to the combination problem, the question of how micro-experiences make a mind. A parent mind is a coherent block of child vibes, the same stuff at a larger scale. The tower of scales, from vibe to particle to cell to organism to world, is renormalization iterated.

8.2. Coherence and the threshold of agency

Coherence measures how tightly a block holds together, the strength of its internal notes combined with the agreement of its tones. As internal coupling rises past a critical value, the block crosses a phase transition. Below it the block merely reacts. Above it a single collective mode governs it, and it acts as a unified agent. So agency above a threshold is the onset of an ordered phase, a computable transition rather than a vague upgrade.

8.3. The self is a Markov blanket

A self is the internal vibes of a coherent block, separated from the external vibes by a blanket of notes, the notes through which the inside senses (fill) and acts (will) on the outside. Statistically this is a Markov blanket, a boundary that renders the internal states conditionally independent of the external. Self and other are therefore not metaphysical absolutes but a structural boundary in the note-field, dynamic and redrawn each beat. What is other can become self by being bound in, and self can become other by dissolving.

A natural minimal picture is a candle flame or a whirlpool in a stream. It is a pattern, not a substance. The matter passing through it is never the same twice, yet it holds a stable shape with a definite boundary, maintained against the flow by drawing in fuel and exporting heat. That is a self in miniature, a coherent pattern with a blanket boundary. The self is the form, not the stuff.

8.4. Mind as the minimization of surprise

A self that persists must keep its blanket intact against a changing environment. The principle that captures this is active inference (Friston, 2010). The self acts and perceives so as to minimize its free energy, its surprise, the mismatch between what it predicts and what it notes. Perception is updating internal tones to fit what is noted. Action is projecting will to change what is noted so it fits the model. Learning is the slow reshaping of note strengths toward a better model. So “minimize suffering” (Section 5) and “minimize free energy” are the same imperative, and it generalizes mere tone-alignment. A self does not only smooth

its tones, it predicts and acts to confirm its predictions, which is why it stays differentiated rather than dissolving into the environment.

8.5. Will, choice, and learning

Base will is a vibe's tone turned outward, mechanical and without purpose. Volition is not primitive. It emerges as base will aggregated over a coherent self and steered by the free-energy gradient. The self projects more strongly in the direction its model predicts will lower its suffering. Purpose is base will plus coherence plus the gradient. Because attention is structural (Section 4), the self cannot steer by instantly re-attending. It steers by where it projects its will and by slowly reshaping its note-structure.

A coherent self does not flip instantly when its inputs change. Its internal vibes must re-reach internal agreement, which takes a number of beats, the settling time of its internal dynamics. That interval, experienced from inside, is deliberation. From outside it is a deterministic relaxation. Both are true. Why it still feels open is computational irreducibility. There is no shortcut to the outcome faster than running the self, so no observer can know the choice before it settles. Learning and karma are the same slow process. Notes between vibes that keep agreeing strengthen, carving the attractor landscape that is memory, and actions leave persistent note-changes that propagate until they balance.

9. Geometry and Space

There is no container. Space is a derived description of the note-structure. Distance is how far apart two vibes are in the web of notes, and the large-scale shape of that web is the geometry of the world.

9.1. Space is note-distance

Two vibes joined by strong notes are close. Weakly joined, they are far. The fundamental metric is the shortest path through the note-structure, weighted so that strong notes are short. This is purely combinatorial, hop counts and integer strengths rather than real coordinates. Closeness is literally strong notes. Space is made of the relations, not a stage they sit on, and smooth geometry is the large-scale limit only. By the principle of no hidden state (Section 2), a vibe holds no coordinates. Its position is its place in the note-structure, computable from local notes by a finite procedure, not a label it carries.

9.2. What geometry, and why

If the mesh is a regular tiling $\{p, q\}$, meaning p -sided cells with q meeting at each corner, one integer inequality fixes its geometry,

$$(p-2)(q-2) < 4 \text{ spherical, } (p-2)(q-2) = 4 \text{ flat, } (p-2)(q-2) > 4 \text{ hyperbolic.}$$

So the question of what geometry the world has reduces to two integers. The model leans hyperbolic, where $(p-2)(q-2) > 4$, because hyperbolic space gives what the framework wants and the flat plane cannot. A vibe can have a small set of immediate notes yet be only a few notes away from vast numbers of other vibes, since the number of vibes within n notes grows exponentially. This matches the eternal-expansion commitment of Section 7. The substrate has unbounded room a few steps out.

Proposition 6 (The exponential tree) Take a regular tree in which every node has q branches, a Bethe lattice. Each node touches only q neighbors, yet the number of nodes within n steps grows like q^n , exponentially. This is hyperbolic geometry in its barest form, small local degree with vast reach a few steps out. Note-distance on such a tree is just the number of branches between two nodes, distance made entirely of connections.

A hyperbolic mesh is a tree wearing a geometric costume, so the nesting hierarchy of selves (Section 8) and the spatial layout are the same object viewed two ways.

9.3. Curvature, gravity, and dimension

Curvature is computed from the note-combinatorics. Positive curvature marks a tightly bound, mesh-like region, a coherent self. Negative curvature marks a sparsely bridged, tree-like region. Gravity is the tendency of vibes to densify their notes and reduce isolation, a flow toward more positive curvature where coherent structures form. Mass concentrates curvature, the discrete echo of mass curving space. A striking discrete bonus is that regular hyperbolic honeycombs exist only up to four dimensions, with nothing in five. If the world wants exponential reach together with regularity, the geometry caps at four dimensions, a candidate, cost-free reason a world of three space dimensions and one of time is special.

9.4. Spacetime

Space alone is the note-metric. Spacetime is that metric together with the causal order of Section 6. The partial order of which update could influence which supplies the light-cone structure, and the note-distance supplies the spatial part. Together they are a causal set, the discrete-spacetime object. The one-note-per-beat bound is the discrete speed of light, and differential beat rates give the time-dilation analogue.

10. Agency and Movement

Because a self is a pattern, not a substance (Section 8), and space is made of notes (Section 9), movement cannot be a lump sliding through a void. It is a pattern re-instantiating itself across a shared, living substrate.

10.1. Movement is pattern propagation

An agent is a self-maintaining coherent pattern of vibes and notes. Its vibes are loci of the substrate, shared with everything else. What travels when it moves is the form, not the material. One step of movement proceeds as follows. At the leading edge the agent projects its will, through an existing note, onto a neighboring vibe, so that vibe's own update is dominated by the agent's wills and it adopts the pattern's tone, and the notes binding it in slowly strengthen. At the trailing edge a vibe is no longer reinforced, its notes to the pattern weaken, and it relaxes back to background. After one beat the pattern lives on a slightly shifted set of vibes. The agent has moved one step.

Proposition 7 (The glider) The glider in Conway's Game of Life is a five-cell pattern that, after four steps, reappears one cell over on the diagonal. No cell ever moves. Cells only switch on and off, yet the glider travels across the grid indefinitely. A vibe-agent moves the same way. The vibes stay put and only re-tone, while the pattern propagates. A ripple on water is the same fact in a continuous medium. The water stays in place and the wave moves through it.

Three consequences follow. The agent is reborn one step over each beat, so over enough beats it is on entirely new substrate, and identity is pattern-continuity rather than substrate-continuity. The pattern shifts by at most one note per beat, the discrete light cone of Section 6.

10.2. Recruiting a vibe is not seizure

The leading-edge move looks like taking over a vibe, but it is gentler and stricter than seizure. Nothing is owned, consumed, or moved. The vibe stays put and changes its own tone, and the agent only contributes terms to that vibe's field. Winning a vibe is winning the arithmetic of its own update, not writing into it. Because attention is structural (Section 4), the agent can only recruit vibes it already notes, and binding becomes stable only as the notes strengthen by their own slow dynamics. So movement is railed. The agent advances along the note-structure, one neighbor at a time. Background vibes are fungible carriers, since a vibe has no individuality beyond its tone and notes, so any suitably placed background vibe can play a role in the pattern.

10.3. Moving through other agents

There is one shared substrate, so other agents are other patterns on the same vibes. Movement is never through empty space. It is through a medium made of experience, much of it organized into other selves. When two agents' blankets sit next to each other, their shared boundary notes link them, so each notes the other and they perceive each other. To advance, a leading edge must win its next vibe, and if that vibe is held by another agent the result is a contest, in which the stronger summed will pins the tone. A meeting can end in several ways. The agents may pass through each other like solitons, scatter like a collision, synchronize and travel together, merge into one higher-coherence self, exchange tone-patterns across the boundary without merging, which is communication, or one may strip the other's substrate until it loses coherence and dies.

10.4. Momentum and mass

Momentum is a persistent directional bias in the will. Inertia is coherence resisting re-instantiation, since a more coherent agent must re-form more internal structure per step and so moves slower for the same push. Inertial mass is, in this reading, coherence. Drag is the cost of recruiting committed vibes, so dense regions are viscous and empty background is frictionless. All of this is discrete, recruit and release, one note per beat. Smooth trajectories with a velocity are the coarse-grained average of the jagged glider-walk, the way fluid flow emerges from molecules.

11. Inner Structure and Matter

This is the frontier of the model. A scalar ternary tone gives a vibe a state but not the inner structure that matter needs, namely spin, chirality, fermions, antiparticles, charge, and the exclusion principle. This section sketches how that inner structure can be built without adding any new primitive. The constructions are grounded in known mathematics but are not yet proven on a hyperbolic mesh, so this section is openly speculative.

11.1. The monism requirement on the inside

Matter cannot be added as a new primitive. By monism (Section 2) there is nothing but vibes, tones, notes, and beats, so any internal degree of freedom, such as spin or charge, must be a derived pattern of a small system of vibes and notes, not a substance inside one vibe. “A vibe carries spin” is shorthand for “a small patch of vibes and notes has a mode that behaves like spin.”

11.2. The spinor as a graded complex of vibes

The cleanest construction is the Kähler-Dirac form, in which a spinor is not a mysterious column of components but a tone on each cell of a small patch, read here as a graded complex of vibes. The vertices are base vibes. The edges are relational vibes, the notes themselves. The faces are higher relational vibes over small loops. A spinor is then a tuple of vibe tones across these levels on one neighborhood, knit together by the discrete exterior derivative, which is just the note-adjacency between a relational vibe and the vibes it relates. Every component is a vibe. The number of components is fixed by the cell combinatorics of one tile-neighborhood, not chosen by hand, which is also how the model would address fermion doubling. The gamma matrices that define a spinor become the operation of raising or lowering the relational order, binding or releasing a relational vibe, both pure note operations.

Proposition 8 (The belt trick) Hold one end of a belt fixed and rotate the buckle one full turn. The belt is twisted and cannot be undone without rotating back. Rotate the buckle two full turns and the twist slides off without rotating the buckle at all. The buckle’s “spin,” its 720-degree periodicity, lives entirely in its connection to its surroundings, not in the buckle itself. A spinor in the mesh is the same. The double-cover sign is a holonomy of the notes tethering a patch to the rest, not a hidden coin inside one vibe.

The complex phase a spinor uses is not primitive. By discreteness it is carried by a finite clock-tone, a $\mathbb{Z}_q$ value on a relational vibe, or by the parity of the beat, advancing one step per beat. This same internal tick is a candidate for mass, the rest-frame oscillation rate. Full exchange statistics, the minus sign that makes matter rigid, would come from braiding two such clusters, which is natural on the two-dimensional hyperbolic mesh.

11.3. Toward charge

The same scaffold extends. An internal charge, such as electric or color charge, is a finite representation carried by the vibe-structure, transforming under a finite internal symmetry of the rule, and its conserved value is the most literal karma the mesh has. Discreteness predicts that the allowed charges should be a small finite set rather than a continuum, the right shape for the particle zoo. A particle, in the end, may be best read as a stable topological pattern, a knot in the tone field, whose inner structure is a topological invariant and whose stability is protected by a winding number, rather than a special fundamental vibe.

12. The Emergence of Physics

The model is experience-first, so physics is not the foundation but the large-scale limit of the note-dynamics.

12.1. Experience is the foundation, not the top

It is tempting to draw reality as a stack with a discrete computational substrate at the bottom and experience emerging at the top. That ordering is backward. Experience is the bottom, the only substance, what everything is made of. The discrete mathematics is a description of its form, layered on top. Read from the bottom, the first layer is experience itself, vibes noting vibes, all the way down, with no non-experiential floor beneath it. The next layer is computational form, a discrete rewriting network of the kind explored by Wolfram (2020), which is a mathematical abstraction of the structure and dynamics of the experiential substrate, not a meaningless machine underneath it. The bits are felt tones. The rewrites are note-dynamics. The computation is experience computing itself. Above that come conserved fields and currents at large scale, and finally observed physics, space and gravity and particles, as seen from far away.

Proposition 9 (Form is not substance) Water flowing in a pipe is described exactly by the Navier-Stokes equations, but water is not made of equations. The equations describe the form of the underlying molecular motion. Temperature is the same, real and useful yet a description of molecular motion, not a separate substance. The discrete mathematics stands to experience as Navier-Stokes stands to molecules, a description of the form of the substrate, layered on top, not the substrate itself.

So the correction to digital physics is this. The universe is not made of meaningless bits with experience added on top. It is made of experience, and the discrete computation is the abstract form read off it. Physics is recovered from experience. Experience is never recovered from physics.

12.2. The candidate correspondences

Each piece of physics is meant to emerge as a large-scale limit. Space and distance emerge as note-distance (Section 9). Spacetime and the light cone emerge as the note-metric together with the causal order (Section 6). Energy emerges as the flux of causal notes across a region, in the manner of Wolfram (2020). Curvature and gravity emerge as the combinatorial curvature of the notes. The Dirac equation and spin emerge as a quantum-walk coin, the graded relational-vibe complex of Section 11, in the manner of the quantum cellular automata of Lent et al. (1993) and Watrous (1995). Conserved currents emerge as invariants of the rule. None of these is yet a derivation. Each is a correspondence whose limit must be proven, a long program. But the shapes line up, and discreteness places the model in the family of digital-physics programs that already exhibit these emergences, such as those of Fredkin (2003) and 't Hooft (2016).

12.3. What is established, and what would distinguish it

The honest summary is that the model is internally coherent and constructively specified, with a clear build target, but it is not yet an empirically tested theory, and its connection to established physics is a program rather than a result. The pieces that follow from the axiom and constraints are derived. The minimal-model choices, namely ternary tone, the eternal-expansion fate, hyperbolic geometry, structural attention, and asynchronous scheduling, are committed working choices with named alternatives held open. The harmony-as-energy and free-energy-as-mind readings are well-motivated candidates. The matter construction is open.

Because the model is fully discrete, it is fully simulable, so it can make coarse, checkable predictions. Agency above a coherence threshold predicts a sharp phase transition at a computable critical coherence, not a

smooth ramp. The settling-time reading of deliberation predicts a measurable relation between a system's integration and its decision latency. The model differs from Integrated Information Theory (Tononi, 2008) in treating integrated information as a coherence diagnostic on an experiential substrate rather than the source of experience, so a case where the two disagree about which system is one mind would discriminate them. A genuinely discrete base predicts a smallest scale and small departures from perfect continuity, the standard signatures shared with causal-set programs (Bombelli et al., 1987).

13. Open Questions and Build Order

Several questions remain genuinely open, and the model's value lies partly in making them precise.

The two fate forks of Section 7 are the deepest. Whether the whole can genuinely forget a distinction, and whether a changeless state can be fully aware, together fix the ending of the cosmology and whether its beginning needed explaining. Both turn on premises the framework derives rather than assumes, so both are open to revision.

The matter construction of Section 11 is the hardest technical problem. Getting exactly the right component count without spurious doublers on a hyperbolic mesh, deriving full exchange statistics, and fixing precisely when a note crystallizes into a relational vibe are all unproven. So is the deepest consistency question, whether the inner spinor and the emergent metric co-emerge consistently rather than in a circle.

The engine of beats, made fully rigorous, requires showing that the vibe population grows without bound under the chosen rule, or, if not, estimating the cycle length. The exact rule remains to be found by searching the critical band for the rule that maximizes coherence or integrated information. And the central program, recovering relativity and quantum mechanics as true coarse-grained limits, is a long undertaking that would move the framework from philosophy toward physics.

A concrete build order organizes the work. First, simulate the minimal model, ternary tones over structural notes with asynchronous updates, and verify harmony descent, the formation of coherent selves, the instability of uniform unity, the deliberation gap, and emergent note-distance. Second, search the critical band for the rule. Third, build the spinor construction on a small tile-neighborhood and check its spectrum. Fourth, derive one physical law as a true limit, beginning with gravity as a combinatorial curvature flow. Fifth, test the discriminating predictions of Section 12. Each step is finite and simulable, which is the practical payoff of insisting that the base be discrete.

14. Conclusion

This paper has set out a discrete model of reality built entirely from experience. From one axiom, that to exist is to experience or be experienced, and two constraints, that the base is finite and that the only entity is the vibe, the model derives a working account of its own dynamics, the energy experience minimizes, the discrete time it runs in, the necessity and character of its own beginning, the unbounded growth that follows, the emergence of selves as self-maintaining patterns, the emergence of space and its hyperbolic shape, the mechanics of how an agent moves through a substrate woven from other agents, and a route toward building matter without any new primitive.

Two commitments give the model its shape. Discreteness keeps it finite and simulable, so every claim becomes a computational experiment and every quantity an exact integer rather than an approximation. Monism keeps it built from one kind of thing, so links, fields, particles, space, and time are all patterns of vibes noting vibes, and the inside of a particle is a small system of vibes rather than a new substance. Read together, these two demands turn a philosophical intuition into a constructive program.

The single thesis is that experience is the foundation, not an emergent extra. The discrete mathematics that describes how the substrate updates is a description of its form, layered on top, in the way that the equations of a fluid describe molecular motion without being what the fluid is made of. Physics is recovered from experience. Experience is not recovered from physics.

The model remains a philosophical and mathematical proposal rather than a tested theory. Its connection to established physics is a program, its account of matter is a frontier, and its deepest cosmological questions are openly unresolved. What it offers is a coherent, discrete, experience-first ontology with a concrete build target, stated precisely enough to be simulated, criticized, and refined.

References

- [1] Lance Pollard. Vibe theory: A philosophical framework centered on experience, 2025. Manuscript.
- [2] David J. Chalmers. Facing up to the problem of consciousness. *Journal of Consciousness Studies*, 2(3):200–219, 1995.
- [3] Galen Strawson. Realistic monism: Why physicalism entails panpsychism. *Journal of Consciousness Studies*, 13(10-11):3–31, 2006.
- [4] Philip Goff. *Consciousness and Fundamental Reality*. Oxford University Press, 2017.
- [5] Bertrand Russell. *The Analysis of Matter*. Kegan Paul, 1927.
- [6] Torin Alter and Yujin Nagasawa. *Consciousness in the Physical World: Perspectives on Russellian Monism*. Oxford University Press, 2015.
- [7] William James. Does consciousness exist? *Journal of Philosophy, Psychology and Scientific Methods*, 1(18):477–491, 1904.
- [8] Alfred North Whitehead. *Process and Reality*. Macmillan, 1929.
- [9] Gottfried Wilhelm Leibniz. *Monadology*. 1714. Various translations available.
- [10] Itay Shani. Cosmopsychism: A holistic approach to the metaphysics of experience. *Philosophical Papers*, 44(3):389–437, 2015.
- [11] Giulio Tononi. Consciousness as integrated information. *Biological Bulletin*, 215(3):216–242, 2008.
- [12] Masafumi Oizumi, Larissa Albantakis, and Giulio Tononi. From the phenomenology to the mechanisms of consciousness: Integrated information theory 3.0. *PLoS Computational Biology*, 10(5):e1003588, 2014.

- [13] Karl Friston. The free-energy principle: A unified brain theory? *Nature Reviews Neuroscience*, 11:127–138, 2010.
- [14] Judea Pearl. *Probabilistic Reasoning in Intelligent Systems*. Morgan Kaufmann, 1988.
- [15] Roger Penrose and Stuart Hameroff. Consciousness in the universe: A review of the 'orch or' theory. *Physics of Life Reviews*, 11(1):39–78, 2014.
- [16] Henry P. Stapp. *Mindful Universe: Quantum Mechanics and the Participating Observer*. Springer, 2007.
- [17] Stephen Wolfram. *A New Kind of Science*. Wolfram Media, 2002.
- [18] Stephen Wolfram. *A Project to Find the Fundamental Theory of Physics*. Wolfram Media, 2020.
- [19] Gerard 't Hooft. *The Cellular Automaton Interpretation of Quantum Mechanics*, volume 185 of *Fundamental Theories of Physics*. Springer, 2016.
- [20] Edward Fredkin. An introduction to digital philosophy. *International Journal of Theoretical Physics*, 42(2):189–247, 2003.
- [21] John Archibald Wheeler. Information, physics, quantum: The search for links. In W. H. Zurek, editor, *Complexity, Entropy and the Physics of Information*, pages 3–28. Addison-Wesley, 1990.
- [22] Seth Lloyd. *Programming the Universe: A Quantum Computer Scientist Takes on the Cosmos*. Knopf, 2006.
- [23] Max Tegmark. *Our Mathematical Universe: My Quest for the Ultimate Nature of Reality*. Knopf, 2014.
- [24] Roy D. Herbert. Chronoflux research initiative, 2024. Zenodo, CC-BY-4.0.
- [25] L. Bombelli, J. Lee, D. Meyer, and R. D. Sorkin. Space-time as a causal set. *Physical Review Letters*, 59(5):521–524, 1987.
- [26] Fay Dowker. Causal sets as discrete spacetime. *Contemporary Physics*, 47(1):1–9, 2006.
- [27] Joe Henson. The causal set approach to quantum gravity. In D. Oriti, editor, *Approaches to Quantum Gravity*, pages 393–413. Cambridge University Press, 2009.
- [28] Lee Smolin. *Three Roads to Quantum Gravity*. Basic Books, 2001.
- [29] Carlo Rovelli. *Quantum Gravity*. Cambridge University Press, 2004.
- [30] Pablo Arrighi and Jonathan Grattage. A quantum game of life. In *Proceedings of QPL 2011*, pages 31–42, 2012.
- [31] C. S. Lent, P. D. Tougaw, W. Porod, and G. H. Bernstein. Quantum cellular automata. *Nanotechnology*, 4(1):49–57, 1993.
- [32] John Watrous. On one-dimensional quantum cellular automata. In *Proceedings of the 36th Annual Symposium on Foundations of Computer Science*, pages 528–537, 1995.

- [33] Carlo Rovelli. Relational quantum mechanics. *International Journal of Theoretical Physics*, 35(8):1637–1678, 1996.
- [34] Christopher A. Fuchs, N. David Mermin, and Rüdiger Schack. An introduction to qbism with an application to the locality of quantum mechanics. *American Journal of Physics*, 82(8):749–754, 2014.
- [35] Ernst Ising. Beitrag zur theorie des ferromagnetismus. *Zeitschrift für Physik*, 31:253–258, 1925.
- [36] J. J. Hopfield. Neural networks and physical systems with emergent collective computational abilities. *Proceedings of the National Academy of Sciences*, 79:2554–2558, 1982.
- [37] Roy J. Glauber. Time-dependent statistics of the ising model. *Journal of Mathematical Physics*, 4:294–307, 1963.
- [38] M. Blume. Theory of the first-order magnetic phase change in UO_2 . *Physical Review*, 141:517–524, 1966.
- [39] H. W. Capel. On the possibility of first-order phase transitions in ising systems of triplet ions with zero-field splitting. *Physica*, 32:966–988, 1966.
- [40] R. B. Potts. Some generalized order-disorder transformations. *Proceedings of the Cambridge Philosophical Society*, 48:106–109, 1952.
- [41] Henning S. Mortveit and Christian Reidys. *An Introduction to Sequential Dynamical Systems*. Springer, 2007.
- [42] Stuart A. Kauffman. *The Origins of Order: Self-Organization and Selection in Evolution*. Oxford University Press, 1993.
- [43] Yoshiki Kuramoto. *Chemical Oscillations, Waves, and Turbulence*. Springer, 1984.
- [44] Leslie Lamport. Time, clocks, and the ordering of events in a distributed system. *Communications of the ACM*, 21(7):558–565, 1978.
- [45] Elliott H. Lieb and Derek W. Robinson. The finite group velocity of quantum spin systems. *Communications in Mathematical Physics*, 28:251–257, 1972.
- [46] Jerry L. Potter. *Associative Computing: A Programming Paradigm for Massively Parallel Computers*. Plenum Press, 1992.
- [47] Donald O. Hebb. *The Organization of Behavior*. Wiley, 1949.
- [48] Thomas Schreiber. Measuring information transfer. *Physical Review Letters*, 85:461–464, 2000.
- [49] Jakob Hansen and Robert Ghrist. Toward a spectral theory of cellular sheaves. *Journal of Applied and Computational Topology*, 3:315–358, 2019.
- [50] Robin Forman. Bochner’s method for cell complexes and combinatorial ricci curvature. *Discrete and Computational Geometry*, 29:323–374, 2003.

- [51] Yann Ollivier. Ricci curvature of markov chains on metric spaces. *Journal of Functional Analysis*, 256:810–864, 2009.
- [52] M. E. J. Newman and M. Girvan. Finding and evaluating community structure in networks. *Physical Review E*, 69:026113, 2004.
- [53] Maurice Margenstern. *Cellular Automata in Hyperbolic Spaces, Volume 1: Theory*. Old City Publishing, 2007.
- [54] Maurice Margenstern. *Cellular Automata in Hyperbolic Spaces, Volume 2: Implementation and Computations*, volume 2. Archives Contemporaines, 2013.
- [55] Leo P. Kadanoff. Scaling laws for ising models near t_c . *Physics*, 2:263–272, 1966.
- [56] Kenneth G. Wilson. The renormalization group: Critical phenomena and the kondo problem. *Reviews of Modern Physics*, 47:773–840, 1975.
- [57] F. William Lawvere. Diagonal arguments and cartesian closed categories. In *Category Theory, Homology Theory and their Applications II*, pages 134–145. Springer, 1969.
- [58] Brendan Fong and David I. Spivak. *Seven Sketches in Compositionality: An Invitation to Applied Category Theory*. Cambridge University Press, 2019.
- [59] Bob Coecke and Ross Duncan. Interacting quantum observables: Categorical algebra and diagrammatics. *New Journal of Physics*, 13:043016, 2011.
- [60] Roger Penrose. Angular momentum: An approach to combinatorial space-time. In T. Bastin, editor, *Quantum Theory and Beyond*, pages 151–180. Cambridge University Press, 1971.
- [61] Erich Kähler. Der innere differentialkalkül. *Rendiconti di Matematica*, 21:425–523, 1962.
- [62] John Kogut and Leonard Susskind. Hamiltonian formulation of wilson’s lattice gauge theories. *Physical Review D*, 11:395–408, 1975.
- [63] H. B. Nielsen and M. Ninomiya. Absence of neutrinos on a lattice. *Nuclear Physics B*, 185:20–40, 1981.
- [64] David Hestenes. *Space-Time Algebra*. Gordon and Breach, 1966.
- [65] Mathieu Desbrun, Anil N. Hirani, Melvin Leok, and Jerrold E. Marsden. Discrete exterior calculus, 2005. arXiv:math/0508341.
- [66] Juan Maldacena. The large n limit of superconformal field theories and supergravity. *Advances in Theoretical and Mathematical Physics*, 2:231–252, 1997.
- [67] Leonard Susskind. The world as a hologram. *Journal of Mathematical Physics*, 36(11):6377–6396, 1995.
- [68] Edward Witten. Anti de sitter space and holography. *Advances in Theoretical and Mathematical Physics*, 2:253–291, 1998.

- [69] Fernando Pastawski, Beni Yoshida, Daniel Harlow, and John Preskill. Holographic quantum error-correcting codes: Toy models for the bulk/boundary correspondence. *Journal of High Energy Physics*, 2015(6):149, 2015.
- [70] F. Drewes, A. Habel, and H. J. Kreowski. Hyperedge replacement graph grammars. In *Handbook of Graph Grammars and Computing by Graph Transformation*, volume 1, pages 95–162. World Scientific, 2002.
- [71] H. Ehrig, R. Heckel, M. Korff, M. Löwe, L. Ribeiro, A. Wagner, and A. Corradini. Algebraic approaches to graph transformation. In *Handbook of Graph Grammars and Computing by Graph Transformation*, volume 1, pages 247–312. World Scientific, 1997.
- [72] Greg Turk. Generating textures on arbitrary surfaces using reaction-diffusion. *ACM SIGGRAPH Computer Graphics*, 25(4):289–298, 1991.
- [73] Andrew Witkin, Kurt Fleischer, and Alan Barr. Energy constraints on parameterized models. *ACM SIGGRAPH Computer Graphics*, 21(4):225–232, 1987.
- [74] Francis H. Cook. *Hua-yen Buddhism: The Jewel Net of Indra*. Pennsylvania State University Press, 1977.
- [75] Jaideva Singh. *Spanda-Karikas: The Divine Creative Pulsation*. Motilal Banarsidass, 1980.